



Application of Physics-Informed Neural Networks with Topological Validation on Gene Regulatory Networks

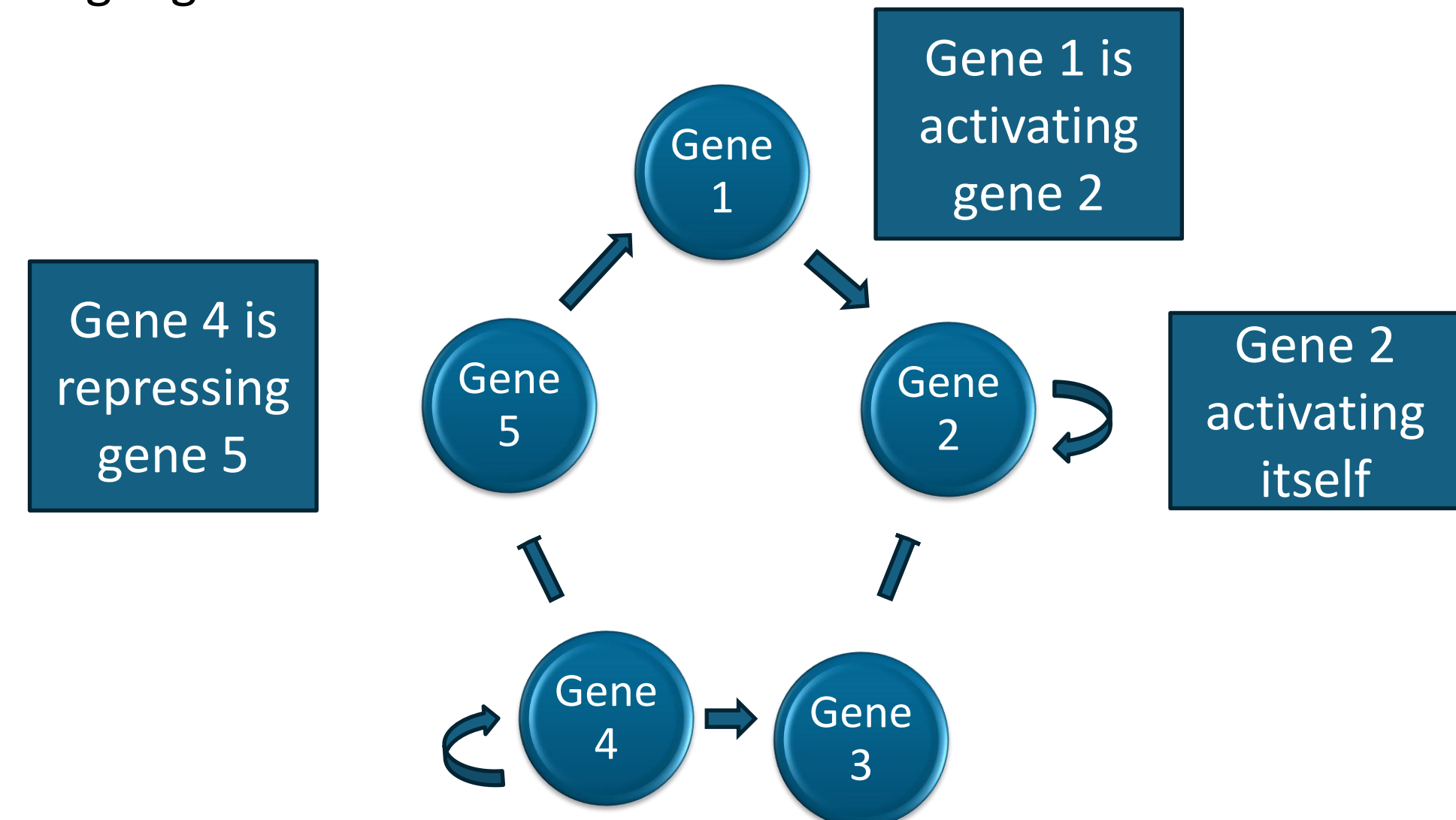


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Gene Regulatory Networks (GRN)

GRNs represent complex interactions among genes and their regulatory mechanisms. Some genes activate, while others repress the transcription of target genes.



Let $x_i(t) \in \mathbb{R}_{\geq 0}$ denote the concentration of protein i at time t . The dynamics are governed by

$$\frac{dx_i}{dt} = -\gamma_i x_i + \sum_{j=1}^m s_{ij} H(\theta_{ij} - x_j),$$

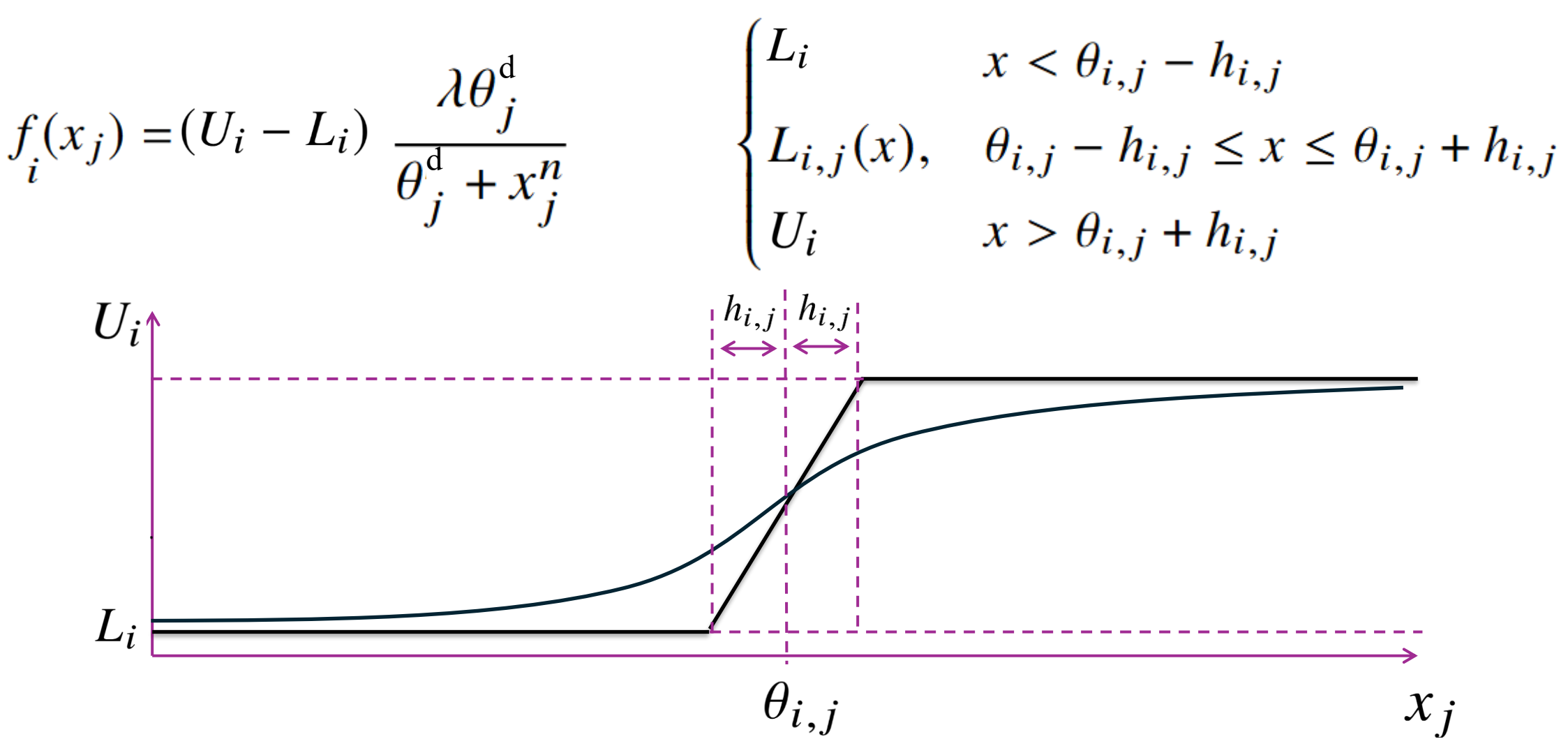
where γ_i is the degradation rate, s_{ij} encodes the sign and strength of regulation from gene j to gene i , θ_{ij} are threshold concentrations, and $H(\cdot)$ is the Heaviside function:

$$H(z) = \begin{cases} 1, & z > 0, \\ 0, & z \leq 0. \end{cases}$$

The Heaviside function can be approximated by 2 smoother/less steep functions:

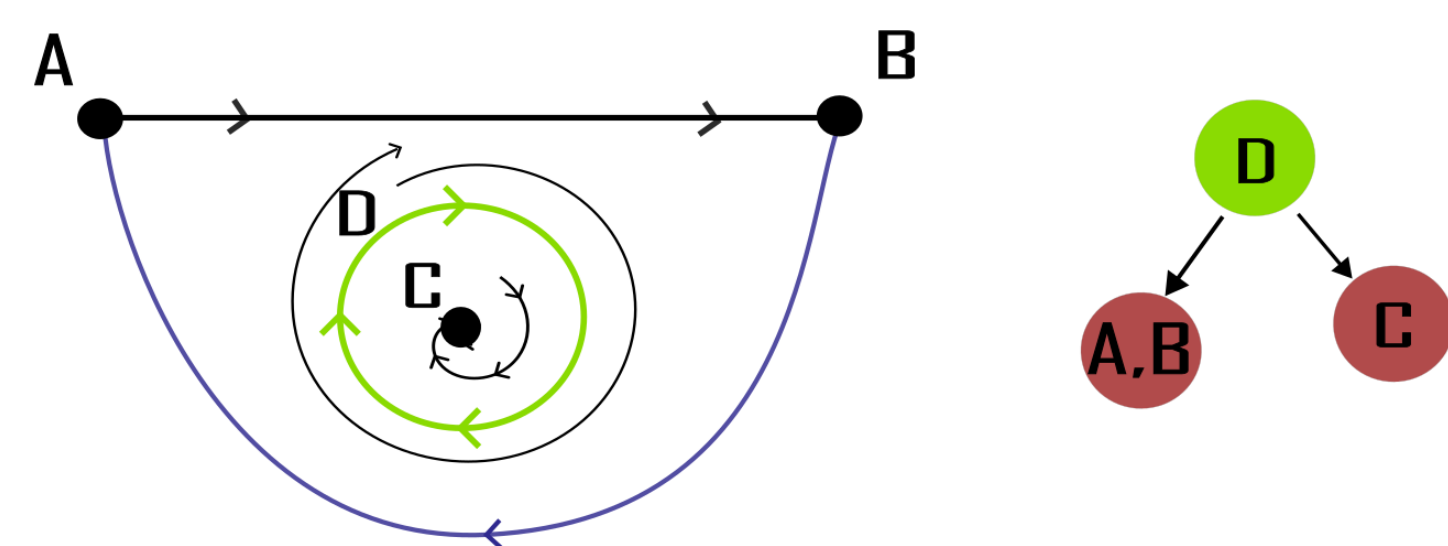
Hill function

Ramp function



Morse Graph:

Let X be a compact metric space and $\varphi: \mathbb{R}_{\geq 0} \times X \rightarrow X$ a continuous semiflow. A family $\mathcal{M} = \{M_\alpha\}_{\alpha \in A}$ of nonempty, pairwise disjoint, compact, invariant sets is a **Morse decomposition** of X if there is a strict partial order $<$ on A such that for every $x \in X$ whose full orbit is defined forward and backward, either $x \in \bigcup_{\alpha} M_\alpha$ or there exist indices $\alpha \leq \beta$ with $\omega(x) \subset M_\alpha$, $\alpha \leq \beta$, $M_\beta \subset \alpha(x)$, where $\omega(x)$ and $\alpha(x)$ are the omega and alpha limit sets of x . All recurrent dynamics is contained in the M_α

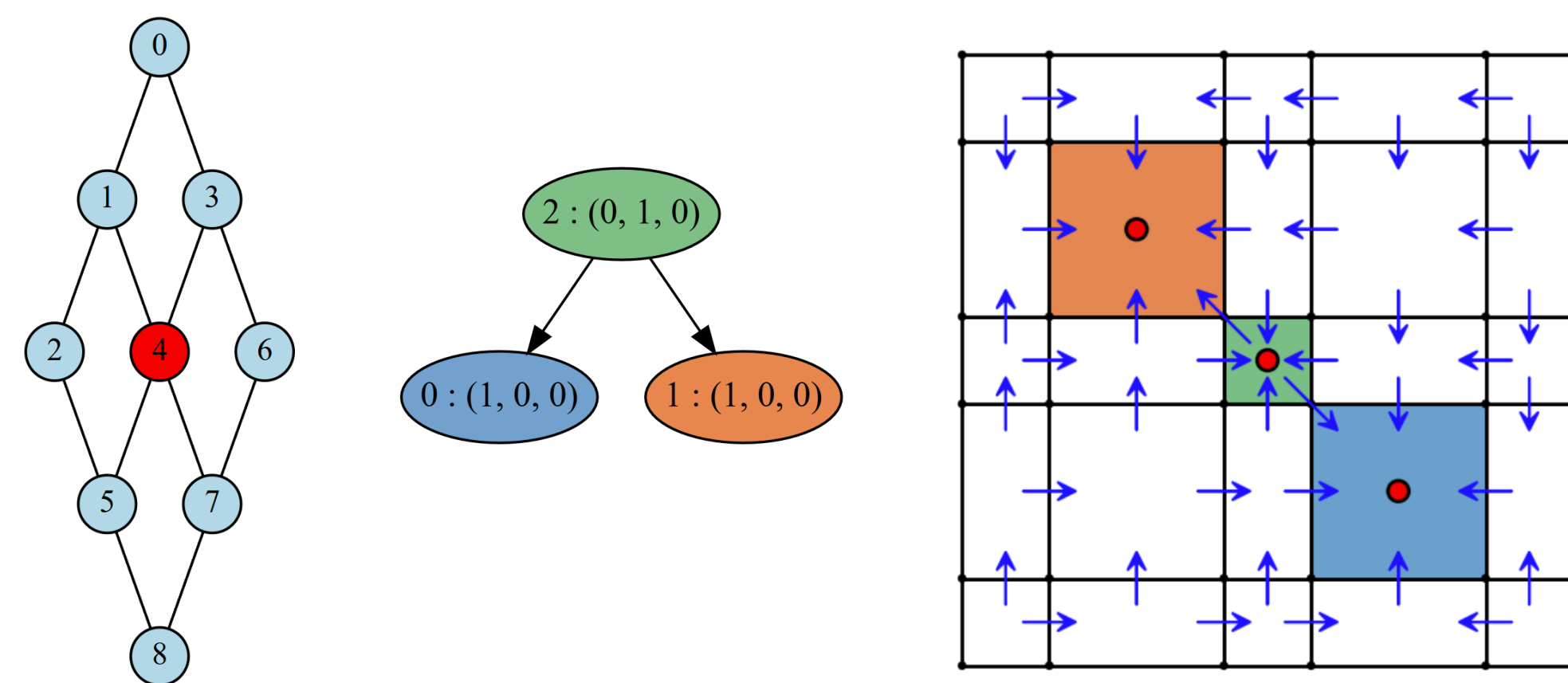


DSGRN:

Dynamic Signatures Generated by Regulatory Networks (DSGRN) is a computational framework designed to analyze the global dynamics of gene regulatory networks without requiring precise parameter values. It partitions the high-dimensional parameter space into distinct regions and assigns each region a qualitative dynamical signature, such as a fixed point, a bistable region, or an oscillation, providing insight into the system's behavior under varying conditions.

Example 1: Toggle switch

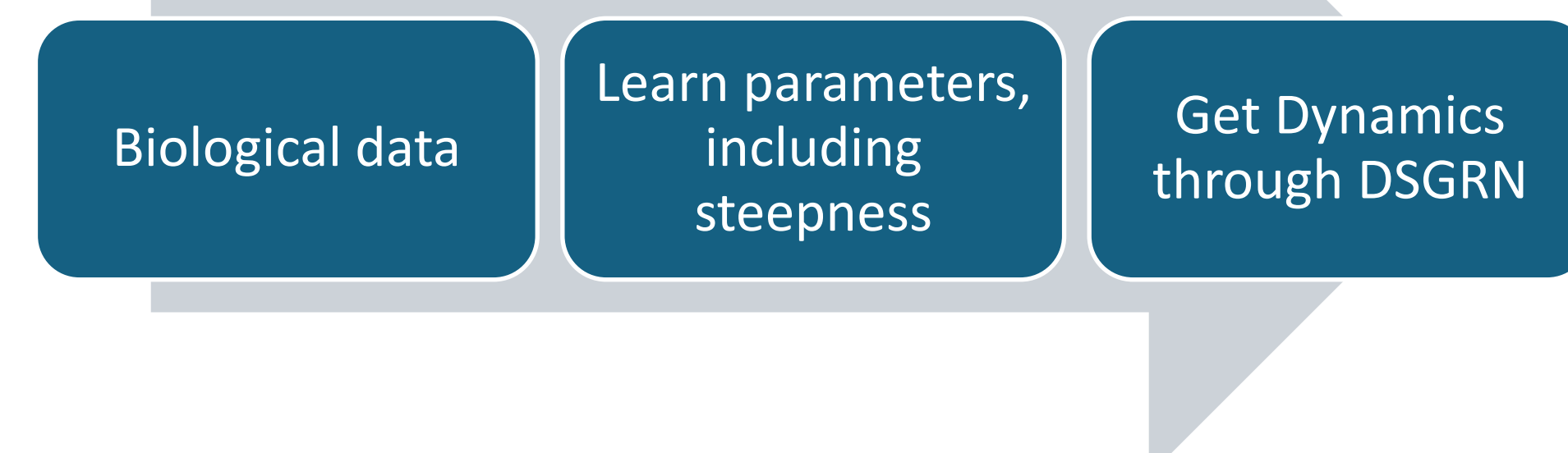
$$\begin{aligned} \dot{x} &= -x + h(y; u_x, l_x, \theta_x, d_x) \\ \dot{y} &= -y + h(x; u_y, l_y, \theta_y, d_y) \end{aligned}$$



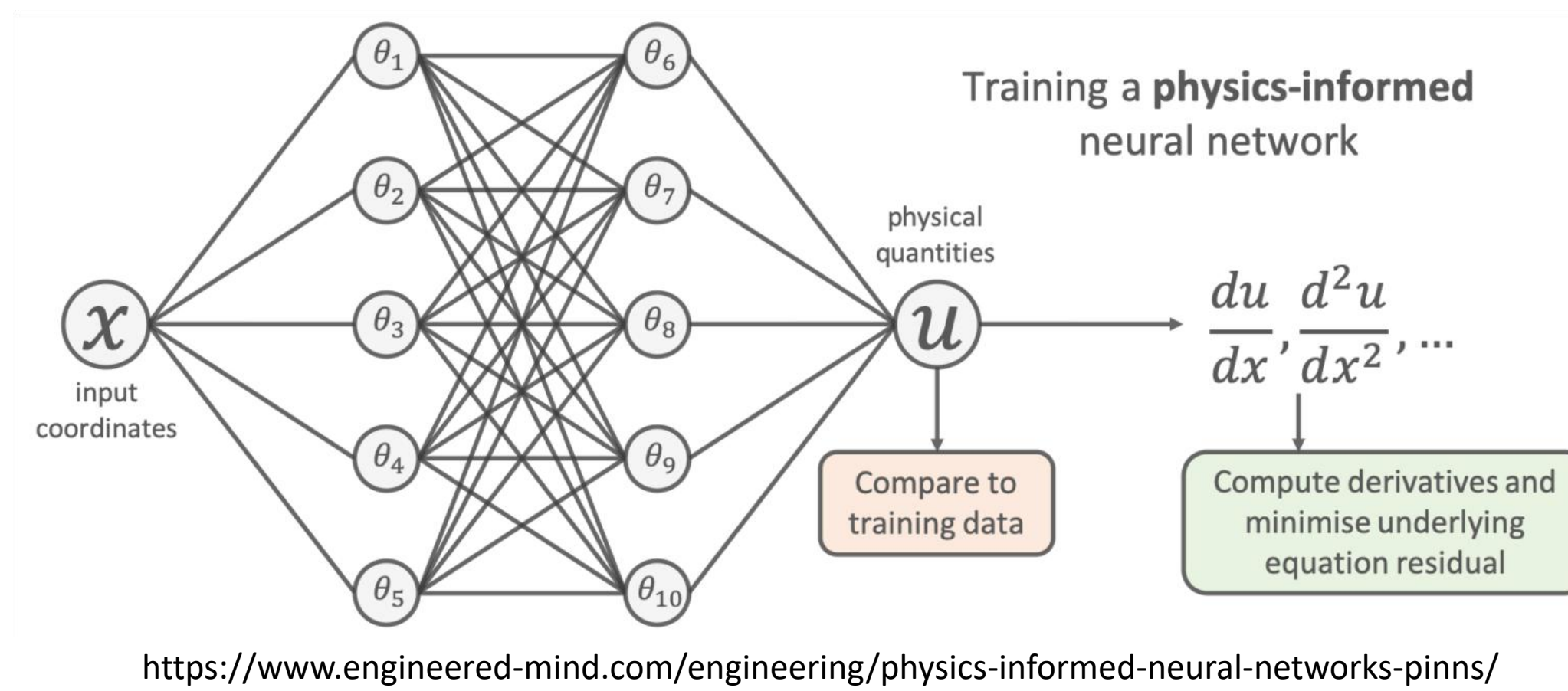
Problem Statement:

For DSGRN to work, it requires that the transition/switching to be defined as a piecewise function. We hereby wish to answer The question:

Given scarce and noisy biological data, can we identify the thresholds L, U , and θ , and detect whether the switching is sufficiently steep?



Our approach: physics-informed neural networks (PINNs)

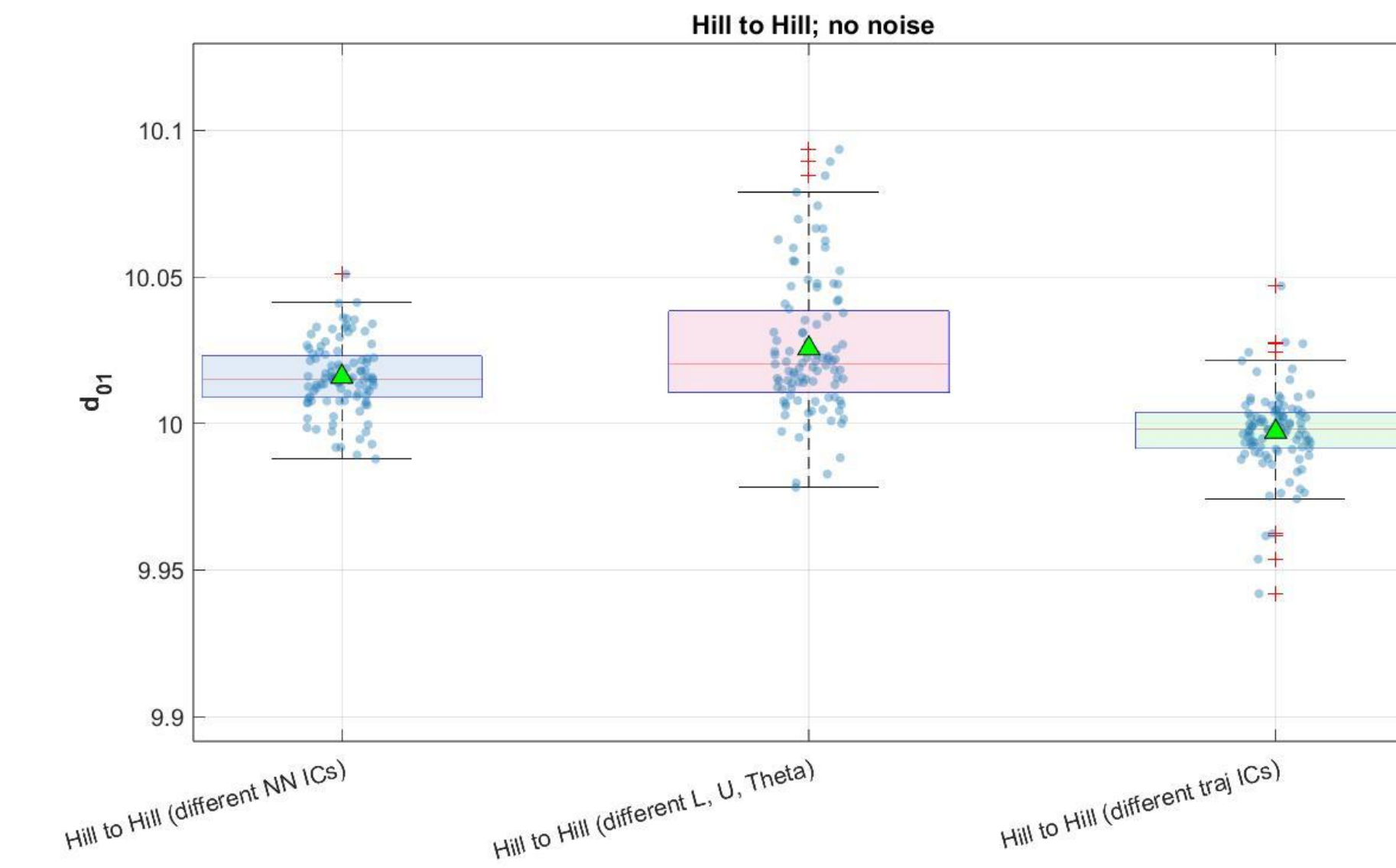


$$\mathcal{L} = \underbrace{\frac{1}{N_d} \sum_{i=1}^{N_d} (y_\theta(x_i) - y_i)^2}_{\text{data term}} + \underbrace{\lambda_f \frac{1}{N_c} \sum_{j=1}^{N_c} (\mathcal{N}[y_\theta](x_j))^2}_{\text{physics residual}} + \underbrace{\lambda_b \frac{1}{N_b} \sum_{k=1}^{N_b} (\mathcal{B}[y_\theta](x_k))^2}_{\text{BC/IC terms}},$$

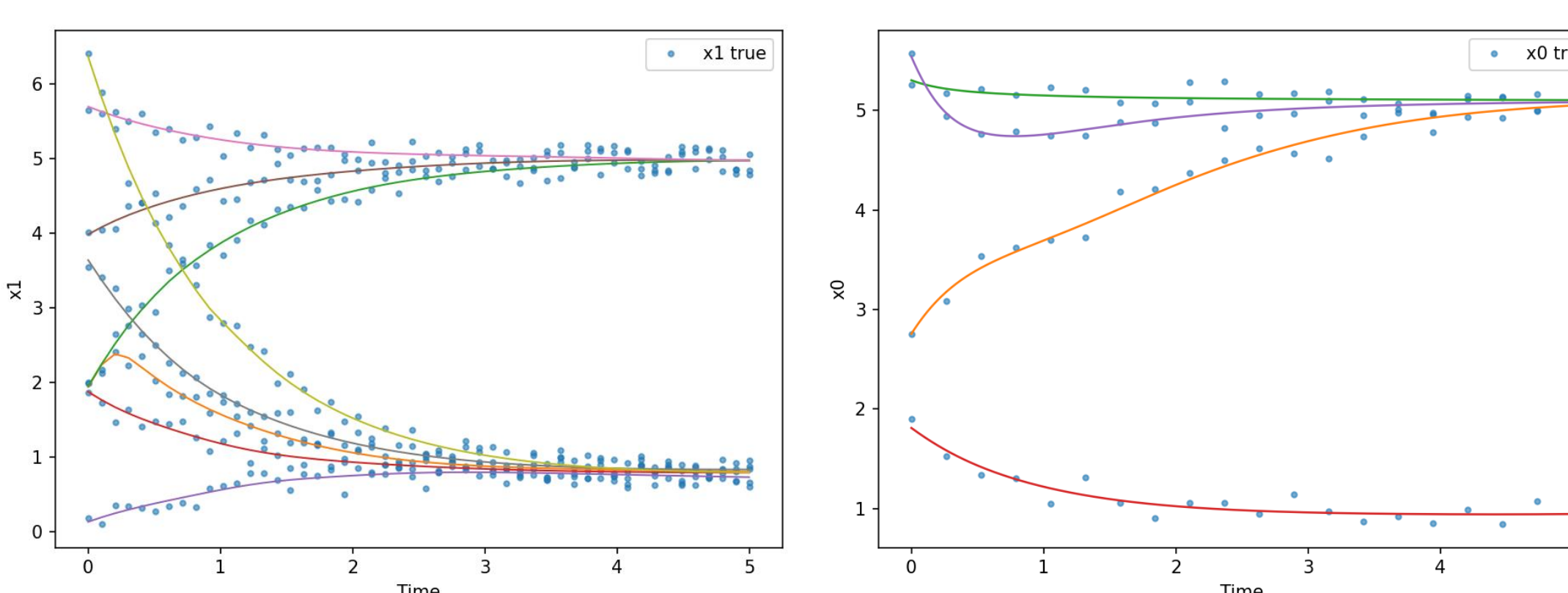
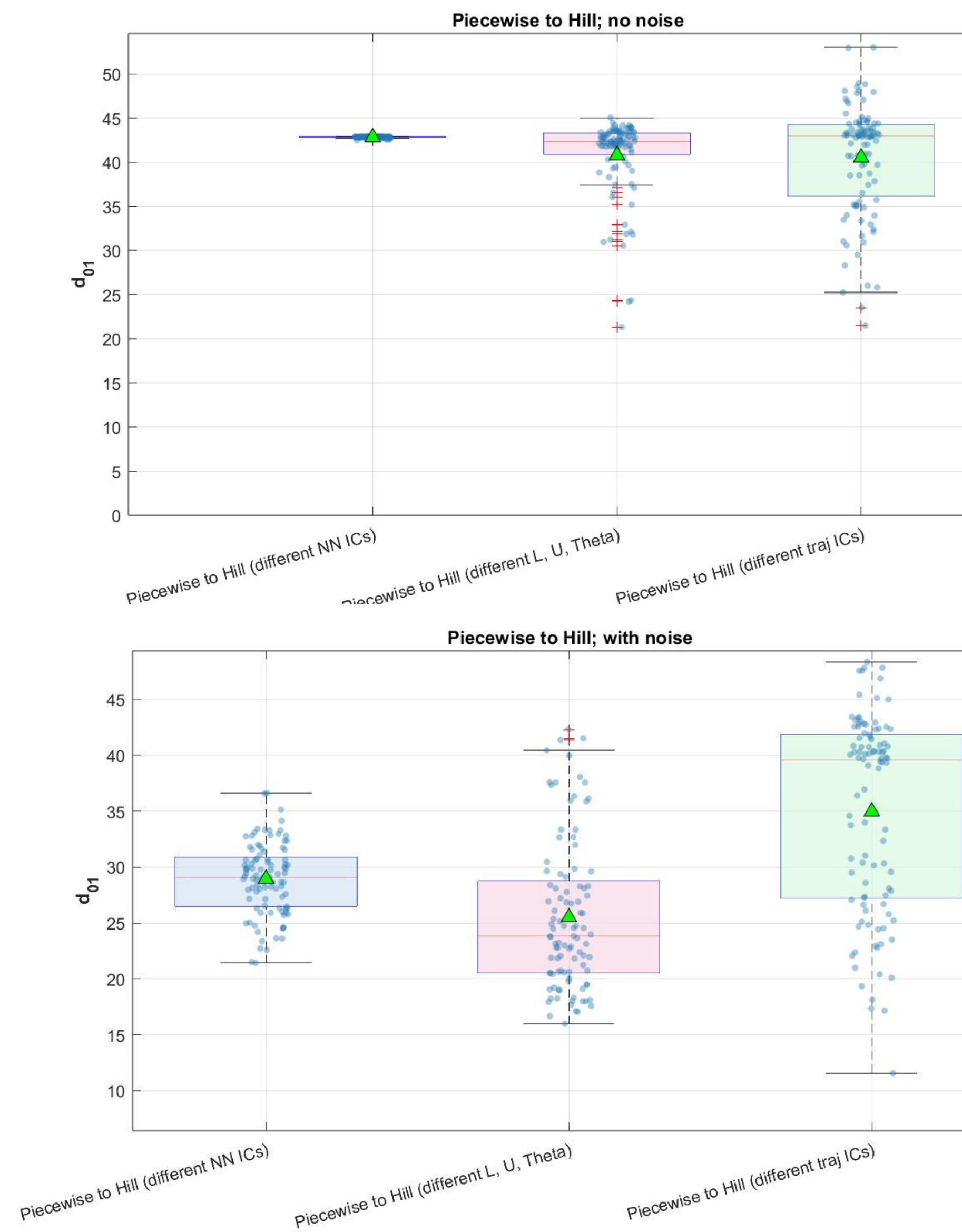
where y_θ is the network output, $\mathcal{N}$ is the differential operator from the governing equation, and $\mathcal{B}$ enforces boundary or initial conditions.

To study steepness in a genetic toggle switch we perform two experiments:

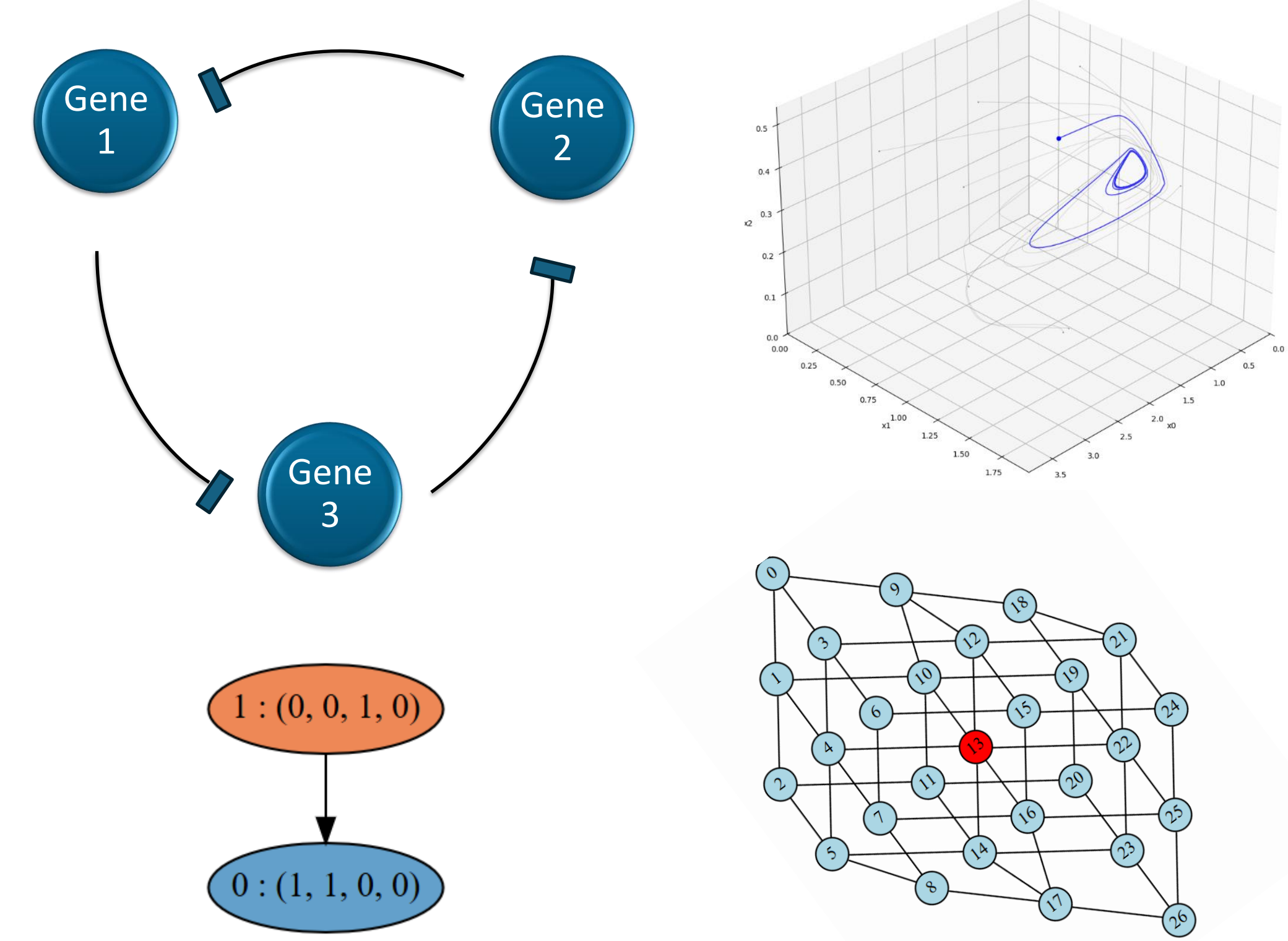
1) **Calibration on Hill data.** We simulate trajectories from an ODE with Hill-type nonlinearities. We vary the neural network initial weights, the Hill parameters L , U , and θ , and the trajectory initial conditions to check that the chosen PINN architecture and hyperparameters are robust.



2) **Validation on piecewise data.** Using the calibrated setup, we fit data generated from piecewise step functions and ask whether the recovered Hill and ramp steepness parameters are consistent with a DSGRN-style switching approximation.



Example 2:



Standard neural networks have tendency to learn low-frequency, smooth components first. Early in training the fit can look almost linear or gently curved, and only later do higher-frequency details appear. We can solve the **spectral bias** problem by introducing feature maps. We provide hereby three types of feature maps:

Fourier Feature map: $A \in \mathbb{R}^{b \times n}$ is a frequency matrix

$$\gamma^1(z) = [\cos(Az), \sin(Az)]$$

Radial Basis Function: c and σ are the centers and standard deviation respectively

$$\gamma^2(z; c, \sigma) = \exp\left(-\frac{\|z - c\|^2}{2\sigma^2}\right)$$

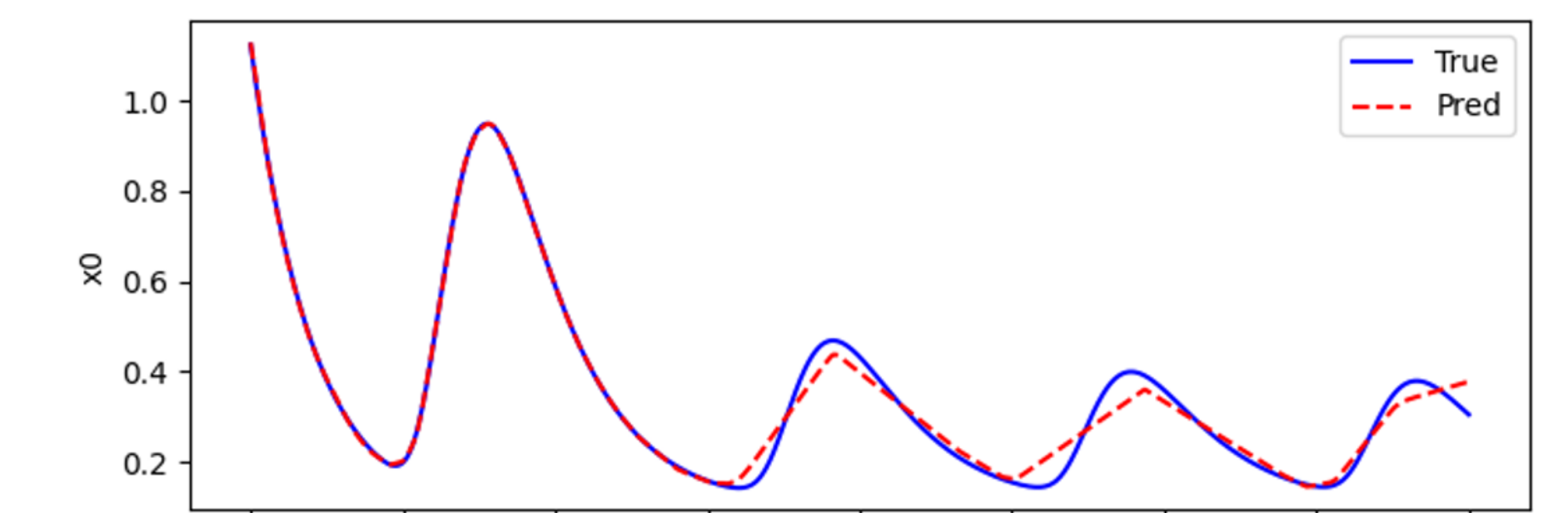
$$[\gamma^2(z; c_0, \sigma), \gamma^2(z; c_1, \sigma), \gamma^2(z; c_2, \sigma), \dots, \gamma^2(z; c_n, \sigma)]$$

Spline map:

$$\gamma^3(x; i) = \begin{cases} \frac{x - t_i}{t_{i+1} - t_i}, & t_i \leq x < t_{i+1}, \\ \frac{t_{i+2} - x}{t_{i+2} - t_{i+1}}, & t_{i+1} \leq x < t_{i+2}, \\ 0, & \text{otherwise.} \end{cases}$$

$$[\gamma^3(z; 0), \gamma^3(z; 1), \gamma^3(z; 2), \dots, \gamma^3(z; n)]$$

No feature map



Fourier Feature map enabled

